Essential Evidence Map

Phage receptor / host-range analysis — *Pseudomonas* phage therapy (iGEM)

2026-07-05

Table of contents

1 What we are doing 1

2 Part A — Our receptors and who shares them 2
- 2.1 Our receptor catalogue . . . 2
- 2.2 The PharBP host list . . . 2
- 2.3 Our receptors × PharBP strains . . . 3
- 2.4 The RBP-side view . . . 3

3 Part B — The folding attempt and why it is a negative result 3
- 3.1 Condensed candidate lists . . . 3
- 3.2 The simulation . . . 6
- 3.3 Control test for ESMFold . . . 7

> **What this document is**
>
> A standalone walkthrough of the key files behind our phage–host analysis: what each is, what it shows, and how they chain together. Figures and table snippets are embedded and numbered. Source files: `code/output/0703_phage_host_matrix/`.

1 What we are doing

We are a phage-therapy project working with several **sequenced *Pseudomonas* isolates** (`9-6-2`, `9-2-1`, `10-10-2`, `8-5-2`, `8-10-2`). The goal is to determine **which surface receptor our phages target** on these bacteria. Our own lytic phages are isolated (plaques) but **not yet sequenced**, so we cannot read their receptor-binding proteins (RBPs) directly.

We therefore bridge indirectly:

1. **Catalogue** each isolate's surface receptors (Section 2.1).
2. Ask which known *Pseudomonas* phages (from the **PharBP** database) infect strains that **share those receptors** (Section 2.2, Section 2.3) — a clue for which phages / RBP classes could target our isolates.

3. **Attempt structural folding** (RBP × receptor) to sharpen the clue (Section 3) — which proved non-informative and is reported as a controlled negative result (Section 3.3).

2 Part A — Our receptors and who shares them

The primary, clue-level evidence. Does not depend on the folding step in Part B.

2.1 Our receptor catalogue

`receptor_by_strain.tsv` organizes all isolates' receptors into **144 unique receptor groups**, marking overlap across isolates. Groups present in **all 5 isolates (core) = 43**; groups present **only in 8-10-2 = 46** — a candidate driver of a difference in host range (Table 1).

Table 1: Receptor catalogue (snippet; 144 groups total, 43 core / 46 unique to 8-10-2).

cluster_rep	category	n_strains	product
BPBBIK_01263	FadL	5 (core)	Long-chain fatty acid transporter
BPBBIK_00419	LptD	5 (core)	LPS-assembly protein LptD
BPBBIK_01078	OmpA	5 (core)	OmpA-like domain-containing protein
NEEDOG_02224	FadL	1 (8-10-2 only)	Long-chain fatty acid transport protein

2.2 The PharBP host list

`pharbp_pseudomonas_hosts.tsv`: from **6,376 *Pseudomonas* host entries** on PharBP there are **43 unique genomes** (54 named strains; the rest are unspecified "*Pseudomonas*" and unusable). *P. putida* KT2440 stands out — many phages (Table 2).

Table 2: PharBP *Pseudomonas* hosts (snippet; 54 named strains / 43 unique genomes).

host	genome	# PharBP RBPs	# phages
P. aeruginosa PAO1	GCF_000006765.1	3106	590
P. putida KT2440	GCF_000007565.2	227	**71**
P. aeruginosa PA14	GCF_030264075.1	130	40
P. syringae	GCF_018394375.1	192	34

2.3 Our receptors × PharBP strains

`receptor_host_presence.tsv` combines Table 1 and Table 2: how **our** receptors are present across the 54 PharBP strains (# host strains matching at 90 / 70 / 50 % identity, best host, etc.). This is the key intermediate for indirectly predicting the RBP range for our isolates (Table 3).

Table 3: Our-receptor presence across PharBP strains (snippet).

our_rep	category	hosts 90 % id	max identity	best host
BPBBIK_01078	OmpA	6	97.7 %	*Pseudomonas putida*
NEEDOG_04082	other-OM	7	98.7 %	*Pseudomonas putida*

Part A takeaway

We now know **the list of our receptors that appear in PharBP, and the hosts that share the same receptors** — the clue-level readout our PI accepted.

2.4 The RBP-side view

The same information from the RBP angle (Figure 2): reference strains from PharBP (left) and the exact RBPs that bind a given strain (right).

3 Part B — The folding attempt and why it is a negative result

Because both candidate lists were large, we condensed them: for RBPs we kept a representative of each highly-similar cluster (>90 %); for receptors we kept only plausible phage targets (removed efflux pumps etc.).

3.1 Condensed candidate lists

`condensed_rbps.tsv` → **114 RBPs**; `condensed_receptors.tsv` → **63 receptors** (Table 4, Table 5).

Table 4: Condensed RBPs (snippet; 114 total).

rbp_id	type	# members	length	host
RBP0001	TF	25	682	*P. putida* KT2440
RBP0002	TSP	20	806	*P. putida* KT2440 / NBRC 14164

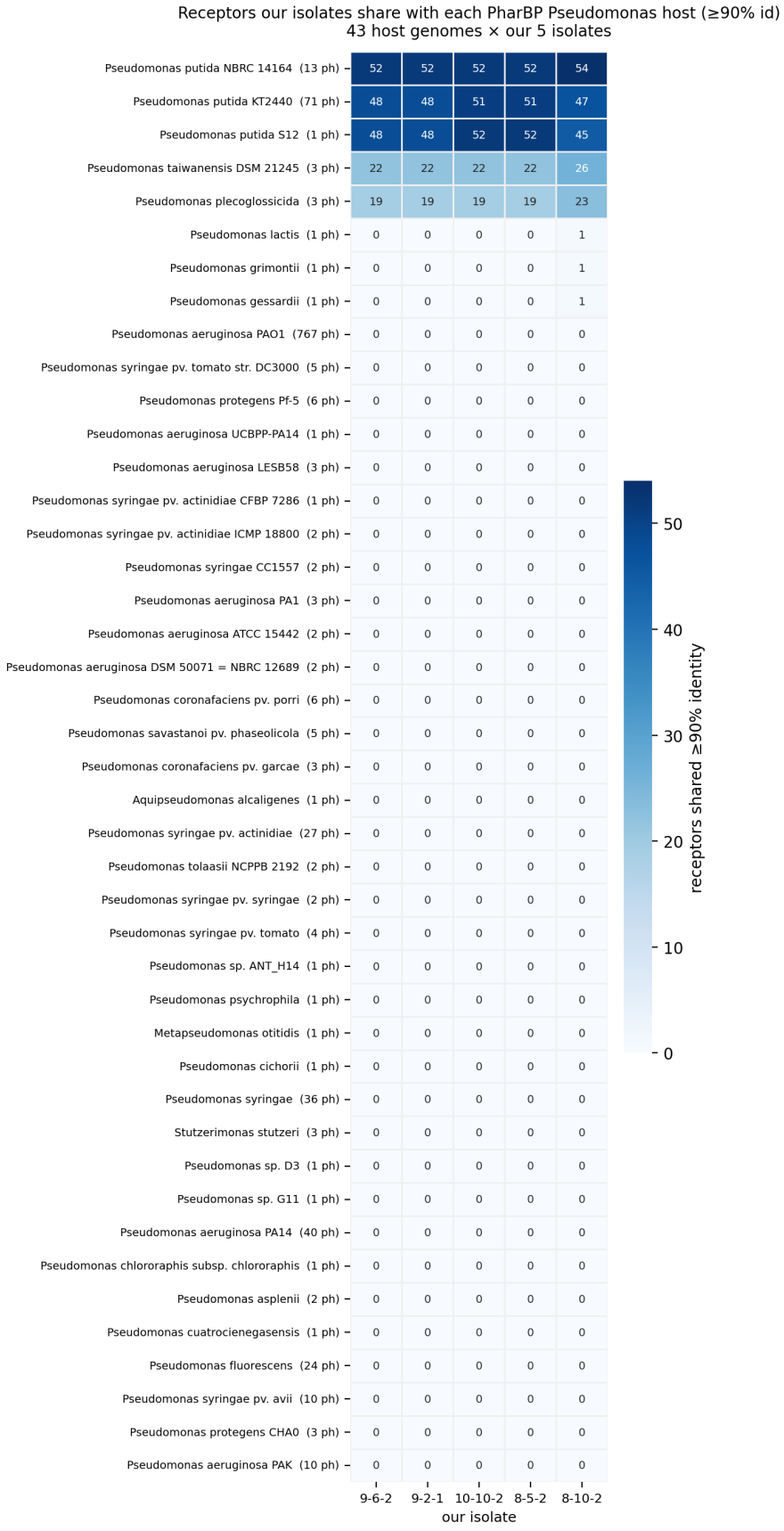
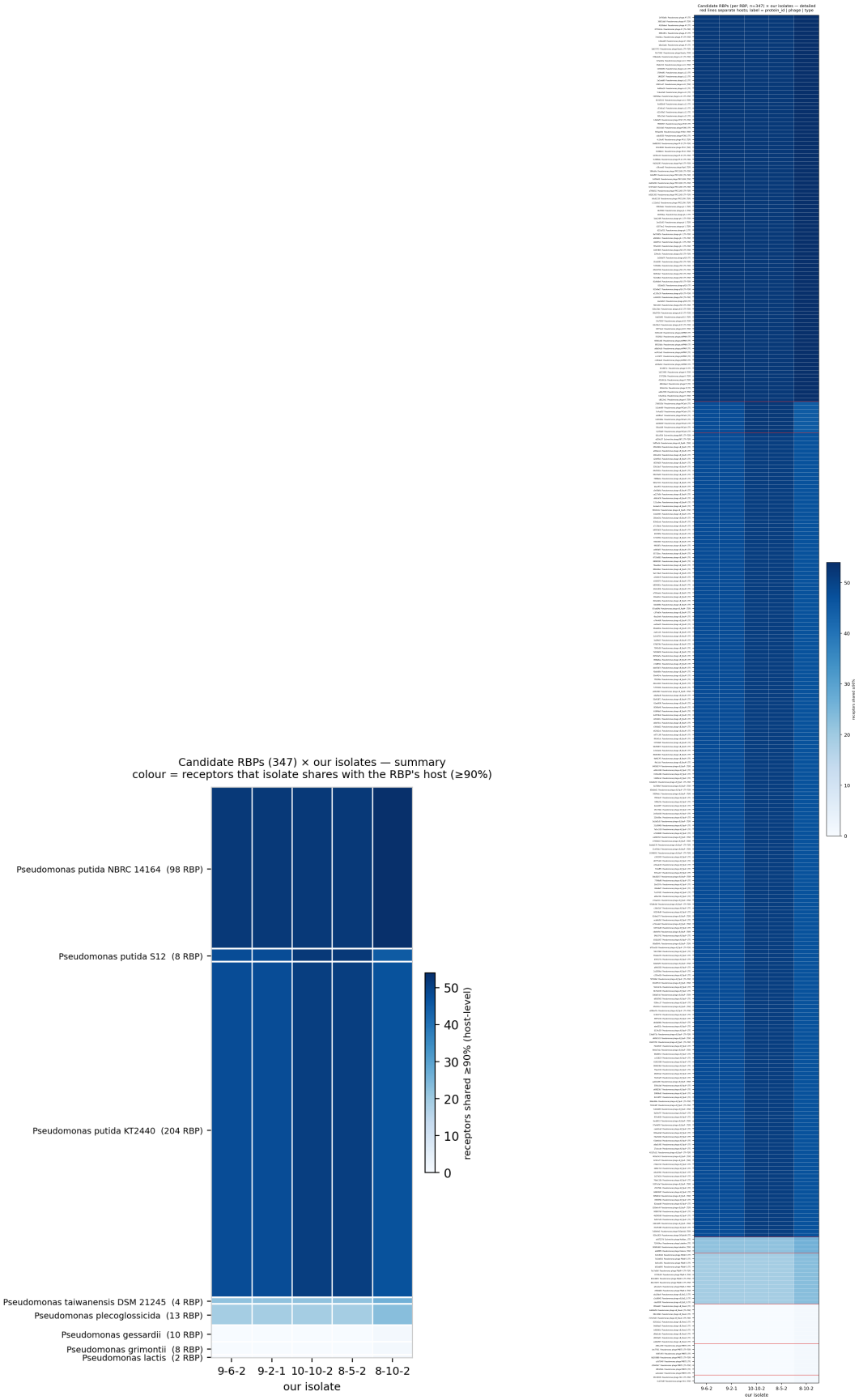


Figure 1: Our receptors × 54 PharBP host strains. Only the *P. putida* group and a few others share a large number of receptors; row labels also give the number of phages infecting each strain. ***P. putida* KT2440** is notable4broad shared-receptor range **and** many infecting phages.



(a) Reference strains (summary)

(b) Exact RBPs per strain (detailed)

Figure 2: RBP × strain views — the other angle on Figure 1.

Table 5: Condensed receptors (snippet; 63 total).

rec_id	category	our strains	product
REC014	FadL	all 5	Long-chain fatty acid transporter
REC006	Omp-porin	all 5	OmpW family protein

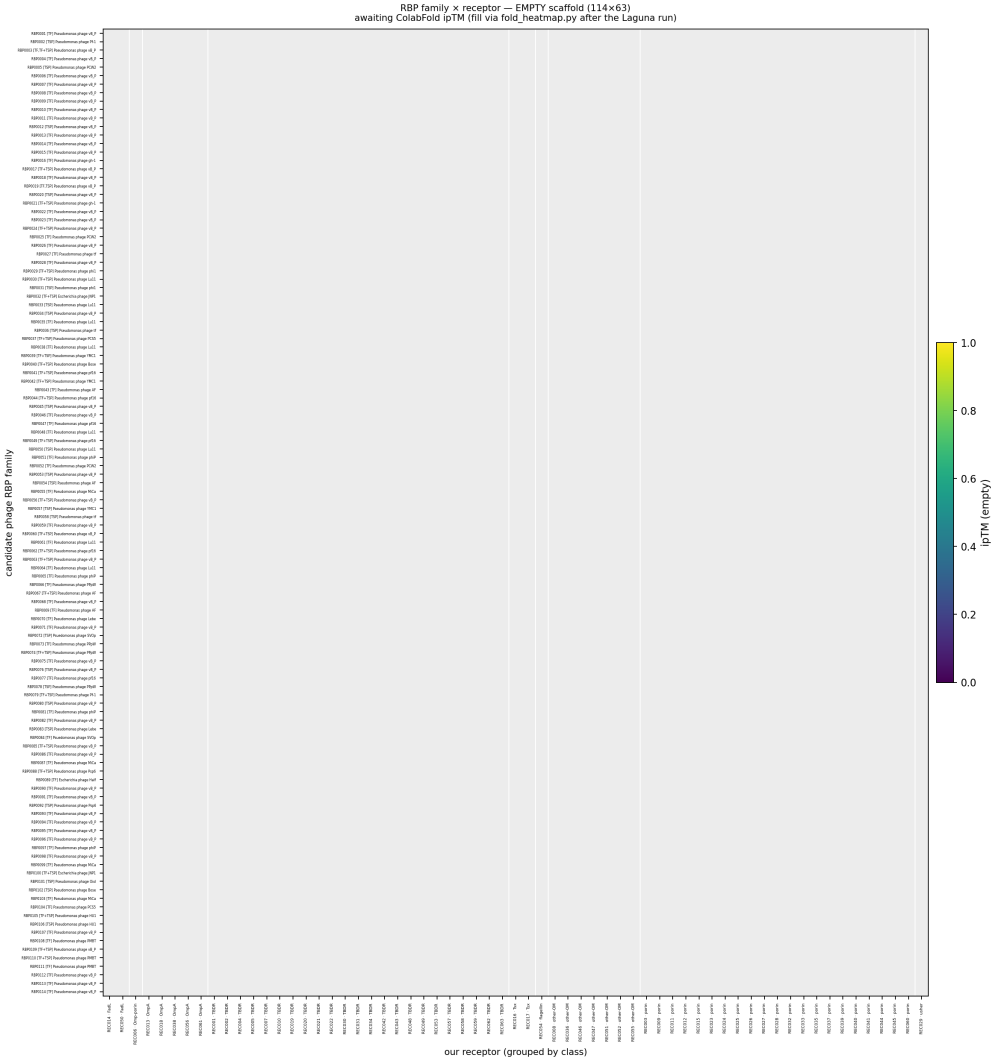


Figure 3: Blank scaffold before any simulation — 114 RBPs × 63 receptors = **7,182** pairs.

3.2 The simulation

`binding_matrix.csv` holds all **7,182** pairs predicted by ESMFold (6,358 folded OK + 824 skipped for length). `iptm` = interaction/binding score; `ptm` = folding quality; `status` = skipped-or-not (Table 6). Large biases follow.

Table 6: `binding_matrix.csv` (snippet; 7,182 rows).

rbp_id	rec_id	iptm	ptm	status
RBP0001	REC001	0.145	0.524	OK
RBP0001	REC002	0.137	0.534	OK

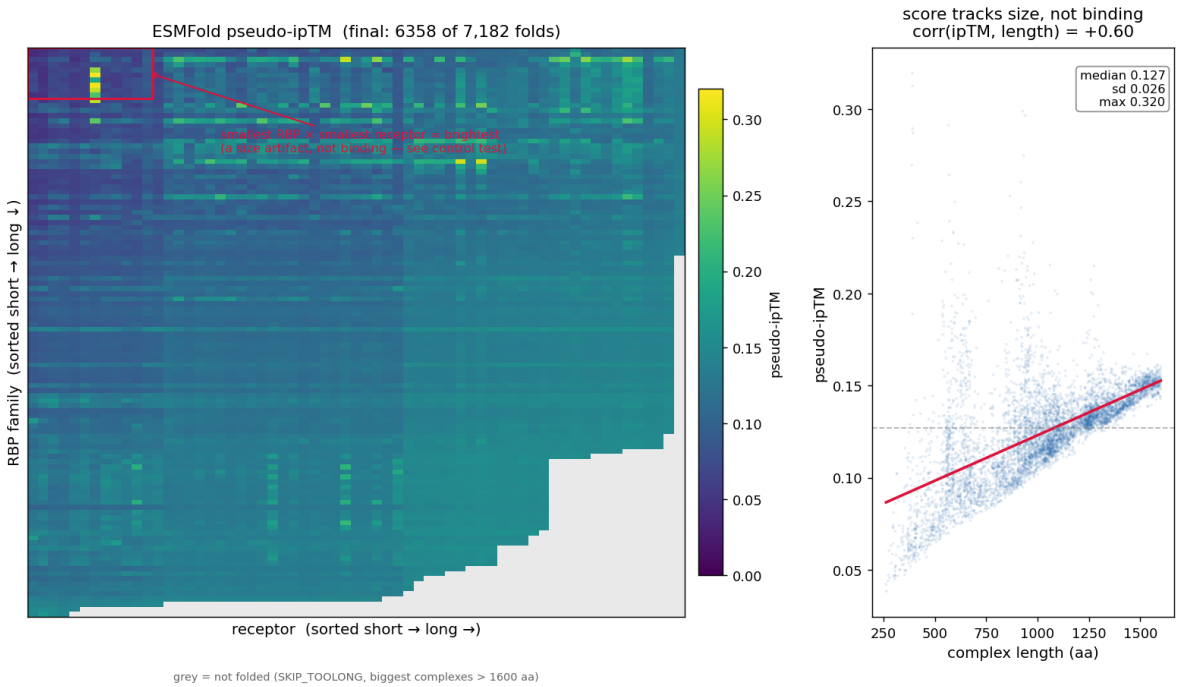


Figure 4: Final ESMFold pseudo-ipTM matrix. **Left:** rows/cols sorted by length — nearly uniform, with the brightest cells at smallest-RBP × smallest-receptor (a size artifact, red box). **Right:** the score correlates with complex length ($\mathbf{r = +0.58}$) — it tracks size, not binding.

3.3 Control test for ESMFold

The matrix gave suspiciously flat scores (everything near 0.13) and the top "hits" tracked protein size, not any specific pairing. A screen is only meaningful if it can separate a real binder from a non-binder, so we tested the method against pairs with **known answers**.

Hypothesis.

- **H1 (method works):** documented binders should score **above** their scrambled/mismatched controls and above the screen background.
- **H0 (null):** the score reflects generic foldability/size, so real scrambled background.

The three positive controls — experimentally documented pairs, prioritizing solved co-structures (ground truth) plus one in our own organism; sequences from UniProt/NCBI (Table 7).

Table 7: Positive controls and provenance.

positive pair	RBP accession	receptor accession	evidence
T5 **pb5** × *E. coli* **FhuA**	P23207	P06971	cryo-EM co-structure, PDB **8A8C**
gpJ (C-term RBD) × *E. coli* **LamB**	P03749	P02943	co-structure, PDB **8XCJ**
DEV **gp54** × *P. aeruginosa* **LptD**	ASZ72261.1	Q9I5U2	genetics + AlphaFold, mBio 2026 (our organism)

For each positive we also folded a **length-matched scrambled** receptor (same residues, shuffled — removes the size confound) and a **mismatched** receptor, through the identical pipeline (Figure 5, Table 8).

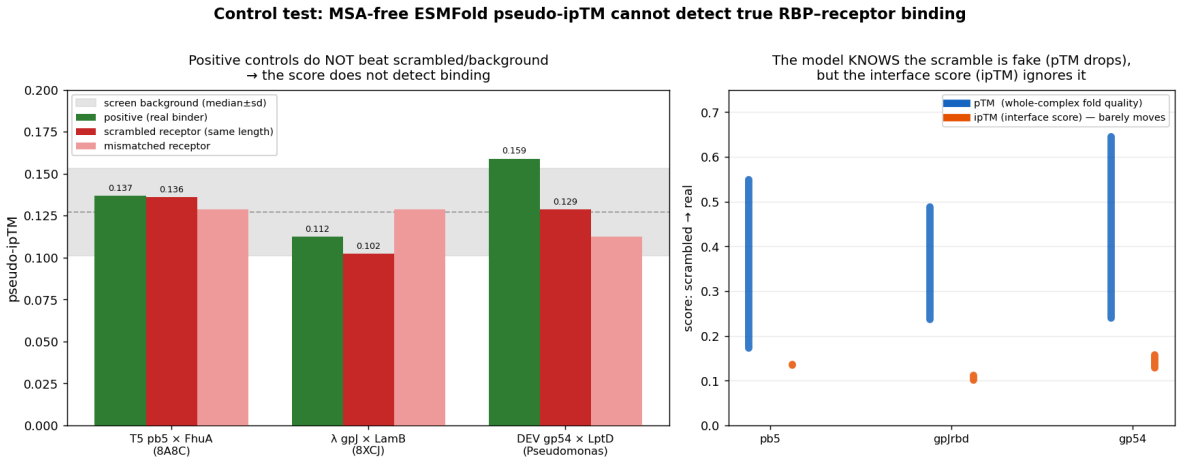


Figure 5: Control test. **Left:** positives (green) do not exceed scrambled (red) or the screen background (grey band). **Right:** the clincher — when the receptor is scrambled, whole-complex **pTM collapses** (the model detects the fake), **but the interface ipTM barely moves** (it is not reading a real interface).

Table 8: Control scores — positives do not separate from negatives or background.

positive pair	real	scrambled	mismatch	background
T5 pb5 × FhuA (8A8C)	0.137	**0.136**	0.129	median 0.127
gpJ × LamB (8XCJ)	0.112	0.102	0.129	(sd 0.026)
DEV gp54 × LptD	0.159	0.129	0.112	

! Conclusion — H0 supported, H1 rejected

Known binders did not separate from controls (the gold-standard pb5 × FhuA differs from its own scrambled receptor by **0.001**). Whole-complex **pTM collapses** on scrambling but the interface **ipTM does not move** → the score is not reading a real interface. **The folding screen is not informative for RBP–receptor binding, and we do not use it as evidence.** Our receptor-repertoire overlap (Part A) remains the primary, clue-level readout and does not depend on this step.

Render: `quarto render 2026-07-05_essential_evidence_map.qmd` → self-contained HTML (images embedded) or `--to pdf`. **Method history:** `2026-07-04_folding_tool_odyssey.md`, `2026-07-04_folding_tool_recommendation.md`, `2026-07-05_esmfold_matrix_note_for_PI_EN.md`, `2026-07-05_phage_host_matrix_reasoning_summary.md`.